1. The number 3.14063 written correct to 3 **decimal** places is

(A) 3.140
(B) 3.141
(C) 3.146
(D) 3.150

2. What percentage of 40 is 8?

(A) 5%
(B) 20%
(C) 32%
(D) 150%

3. Using the distributive property

$$49 \times 17 + 49 \times 3 =$$

(A) $49 \times 20$
(B) $49 + 20$
(C) $52 \times 66$
(D) $52 + 66$

4. The next term in the sequence

1, 6, 13, 22, 33, ______ is

(A) 44
(B) 45
(C) 46
(D) 52

5. 0.45 written as a common fraction, in its simplest form, is

(A) $\frac{9}{20}$

(B) $\frac{4}{5}$

(C) $\frac{9}{10}$

(D) $\frac{5}{4}$

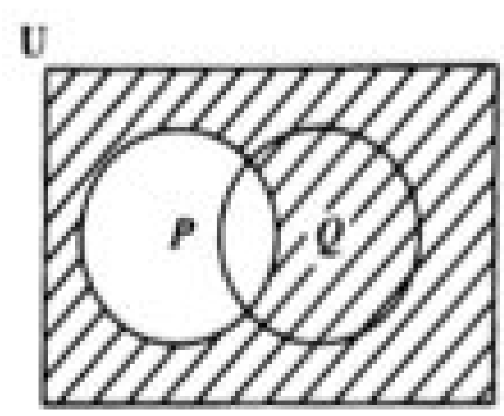
6. A certain amount of money is shared in the ratio 2:3:9. If the difference between the first and second shares is \$40, then the amount of money shared is

(A) \$280
(B) \$360
(C) \$400
(D) \$560

7. Which of the following sets is equivalent to $\{a, b, c, d\}$?

(A) $\{4\}$
(B) $\{a, b, c\}$
(C) $\{p, q, r, s\}$
(D) $\{1, 2, 3, 4, 5\}$

<u>Item 8</u> refers to the following Venn diagram.

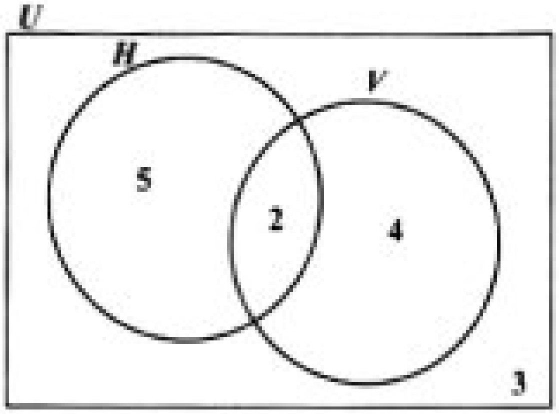



8. The shaded region represents

(A) $P'$
(B) $(P \cup Q)'$
(C) $P \cup Q'$
(D) $Q \cap P'$

GO ON TO THE NEXT PAGE

Item 9 refers to the following Venn diagram.




9. In the Venn diagram

$U$ = {students who play games}
$H$ = {students who play hockey}
$V$ = {students who play volleyball}

The number of students in each set is shown. How many students do NOT play volleyball?

(A) 2
(B) 3
(C) 5
(D) 8

10. If $Q = \{a, b, c\}$ how many subsets can be obtained from the set $Q$?

(A) $2 + 3$
(B) $2 \times 3$
(C) $2^3$
(D) $3^2$

Item 11 refers to the following information on the description of three sets.

$P$ = {prime numbers}
$Q$ = {odd numbers}
$R$ = {even numbers}

11. Which of the following sets is empty?

(A) $P \cap R$
(B) $P \cup Q$
(C) $P \cap Q$
(D) $Q \cap R$

12. If $P$ and $Q$ are two sets where $n(P \cup Q) = 13$, $n(P \cap Q) = 6$ and $n(P) = 9$, then $n(Q) =$

(A) 4
(B) 10
(C) 15
(D) 22

13. A man's taxable income is \$20 000. He pays tax at the rate of 28%. The amount of income tax he pays is

(A) \$3 825
(B) \$4 800
(C) \$5 600
(D) \$7 200

14. A man's basic wage for a 40-hour week is \$160.00. He is paid \$5.00 per hour for overtime. If he works 6 ½ hours overtime in a certain week, his wage for that week is

(A) \$165.00
(B) \$166.50
(C) \$171.50
(D) \$192.50

15. If TT\$6.00 is equivalent to US\$1.00, then TT\$15.00 in US dollars is

(A) \$0.25
(B) \$0.40
(C) \$2.50
(D) \$4.00

16. The annual interest rate on a 15-year mortgage on a house assessed at a value of \$450 000 is 5 cents on every \$1. What is the interest paid on the mortgage for the first year?

(A) \$ 11 750
(B) \$ 22 500
(C) \$107 500
(D) \$117 500

17. Mary invested \$200 for 5 years at 5% per annum. John invested \$500 at the same rate. If they both received the same amount of money in simple interest, for how many years did John invest his money?

(A) 2

(B) $2\frac{1}{2}$

(C) 3

(D) 5

18. A salesman sells a car for \$11 000. If he is paid a commission of 4.5% for the first \$10 000 and 7.5% on the remainder, then the commission he receives is

(A) \$ 495
(B) \$ 525
(C) \$ 825
(D) \$1 320

19. A television set costs \$350 cash. When bought on hire-purchase, a deposit of \$35 is required, followed by 12 monthly payments of \$30. How much is saved by paying cash?

(A) \$10
(B) \$25
(C) \$40
(D) \$45

20. Mr Jones bought a car for \$64 000. The car depreciates by 20% in the first year and 10% in each of the following years. The value of the car at the end of the second year was

(A) \$19 200
(B) \$44 800
(C) \$46 080
(D) \$51 200

21. $\frac{1}{5x} + \frac{2}{3x} =$

(A) $\frac{3}{8x^2}$

(B) $\frac{3}{8x}$

(C) $\frac{13}{15x^2}$

(D) $\frac{13}{15x}$

22. If $5(2x - 1) = 35$, then $x =$

(A) $-4$

(B) $\frac{1}{4}$

(C) $3$

(D) $4$

23. $3x^2 \times 2x^3 =$

(A) $6x^5$
(B) $5x^5$
(C) $6x^6$
(D) $72x^5$

24. Given that $3 * 6 = 12$ and $2 * 5 = 9$, then $a * b$ may be defined as

(A) $4(b - a)$
(B) $a^2 - b$
(C) $6a - b$
(D) $2a + b$

25. When 8 is subtracted from a certain number and the result is multiplied by 3 the answer is 21. What is the **original** number?

(A) 1
(B) 3
(C) 10
(D) 15

26. If $x = -2$, $y = 3$, $r = 2$, then $\left(\frac{x}{y}\right)^r =$

(A) $\frac{-4}{9}$

(B) $\frac{4}{9}$

(C) $\frac{4}{3}$

(D) $\frac{9}{4}$

<u>Items 27–28</u> refer to the following two matrices, $P$ and $Q$.

$$P = \begin{bmatrix} 8 & 6 \\ 7 & 5 \end{bmatrix} \quad Q = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 12 & 3 & 4 & 1 \\ 13 & 14 & 1 & 2 \end{bmatrix}$$

27. What is the order of Matrix $Q$?

(A) $3 \times 4$
(B) $3 \times 2$
(C) $2 \times 3$
(D) $4 \times 3$

28. The determinant of $P$, $|P|$, is

(A) 2
(B) $-2$
(C) $-13$
(D) 26

29. If the vectors **p** and **q** are $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$ respectively, then **p** – 2**q** is

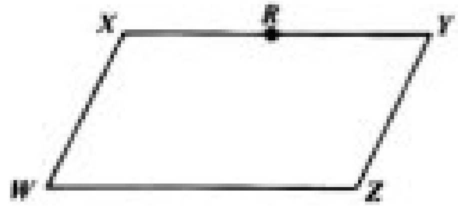
(A) $\begin{bmatrix} 5 \\ -6 \end{bmatrix}$

(B) $\begin{bmatrix} 1 \\ -6 \end{bmatrix}$

(C) $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$

(D) $\begin{bmatrix} 1 \\ 10 \end{bmatrix}$

Item 30 refers to the following parallelogram, $WXYZ$. In the parallelogram, $R$ is the midpoint of $XY$.




30. Based on the information above, $\overrightarrow{ZR}$ =

(A) $\overrightarrow{WX} - \frac{1}{2}\overrightarrow{WZ}$

(B) $\frac{1}{2}\overrightarrow{WZ} - \overrightarrow{WX}$

(C) $\overrightarrow{WX} + \frac{1}{2}\overrightarrow{WZ}$

(D) $\frac{-1}{2}\overrightarrow{WZ} - \overrightarrow{WX}$

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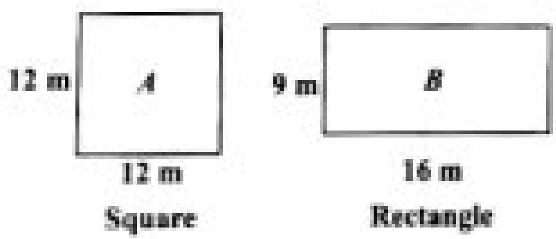
Item 31 refers to the following diagrams of a square and a rectangle.




31. Which of the following statements is true about the perimeter of the square and the rectangle?

(A) Perimeter of $A$ = Perimeter of $B$
(B) Perimeter of $A$ > Perimeter of $B$
(C) Perimeter of $A \geq$ Perimeter of $B$
(D) Perimeter of $A$ < Perimeter of $B$

32. 3 800 millimetres expressed in metres is

(A) 0.38
(B) 3.8
(C) 38
(D) 380

33. The distance around the edge of a circular pond is 88 metres. The radius, in metres, is

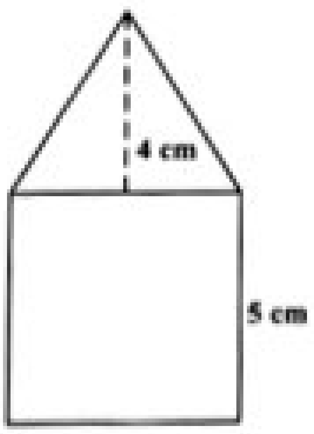
(A) $88\pi$

(B) $176\pi$

(C) $\frac{88}{\pi}$

(D) $\frac{88}{2\pi}$

Item 34 refers to the following figure which shows a triangle resting on a square.




34. The length of one side of the square is 5 cm and the height of the triangle is 4 cm. What is the TOTAL area of the figure, in $cm^2$?

(A) 35
(B) 45
(C) 50
(D) 100

35. The volume, in $cm^3$, of a cube of edge 3 cm is

(A) 9
(B) 18
(C) 27
(D) 54

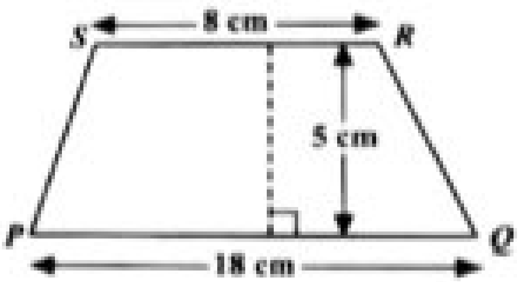
36. The area of a rectangle is 53.6 $cm^2$. If the length is multiplied by four and the width is halved, the area would then be

(A) 26.8 $cm^2$
(B) 53.6 $cm^2$
(C) 107.2 $cm^2$
(D) 214.4 $cm^2$

37. On leaving Trinidad, the time on a pilot's watch was 23:00 h. When he arrived at his destination in the same time zone on the next day, his watch showed 03:00 h. How many hours did the flight take?

(A) 4
(B) 16
(C) 20
(D) 26

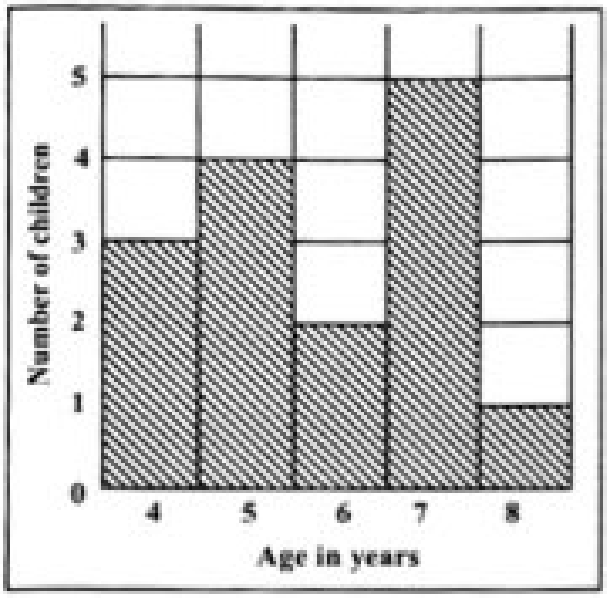
Item 38 refers to the following diagram of a trapezium.




38. The area of the trapezium *PQRS* above is

(A) 45 $cm^2$
(B) 65 $cm^2$
(C) 90 $cm^2$
(D) 130 $cm^2$

Items 39–40 refer to the following histogram which shows the number of children aged 4, 5, 6, 7 and 8 who took part in a survey.




39. What was the modal age?

(A) 5
(B) 6
(C) 7
(D) 8

40. How many children took part in the survey?

(A) 5
(B) 15
(C) 75
(D) 87

41. The median of the numbers

1, 1, 5, 5, 6, 7, 7, 7, 7, 8 is

(A) 7
(B) 6.5
(C) 6
(D) 5.4

42. When three coins are tossed simultaneously, the possible outcomes are {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}, where H represents a head and T represents a tail.

What is the probability of randomly obtaining at LEAST TWO heads?

(A) $\frac{1}{4}$

(B) $\frac{3}{8}$

(C) $\frac{1}{2}$

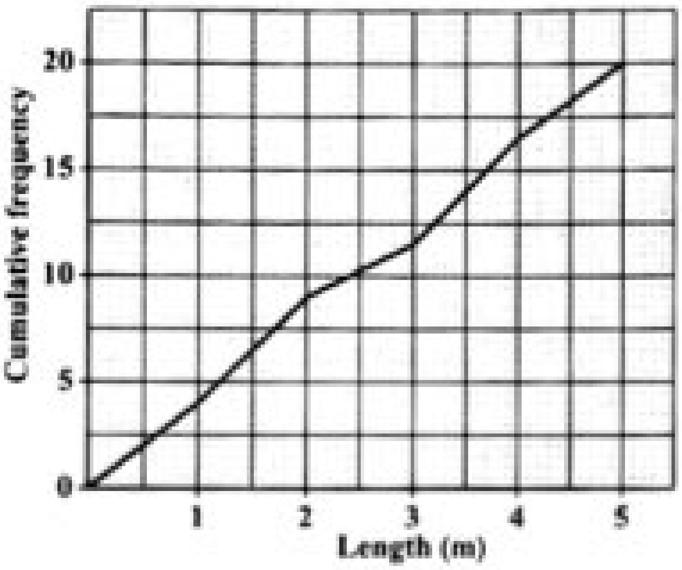
(D) $\frac{2}{3}$

43. Six hundred students write an examination. The probability of a randomly selected student failing the examination is $\frac{1}{5}$.

How many students are expected to **pass**?

(A) 120
(B) 480
(C) 500
(D) 600

Item 44 refers to the following diagram, which shows the cumulative frequency polygon of the lengths, in metres, of 20 fish that are caught by two fishermen.

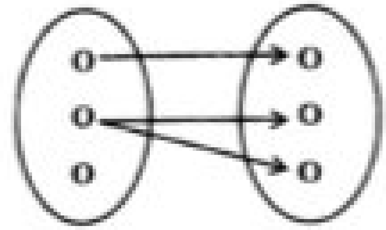



44. The interquartile range of the lengths of the fish is

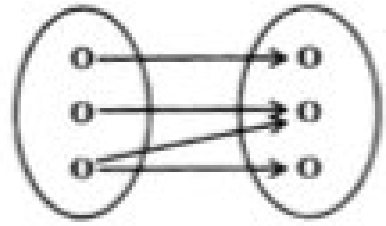
(A) 1.2 m
(B) 2.1 m
(C) 2.5 m
(D) 3.7 m

45. Which of the following diagrams BEST illustrates a function?

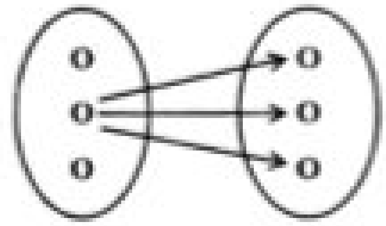
(A)



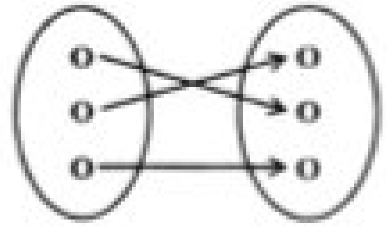
(B)



(C)



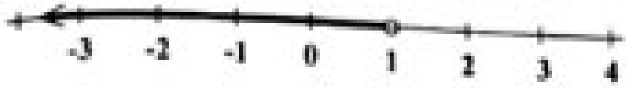
(D)



46. The equation of the line which passes through the point (0, 5) and has a gradient of 4 is

(A) $y = 4x$
(B) $y = 5x$
(C) $y = 4x + 5$
(D) $y = 5x + 4$

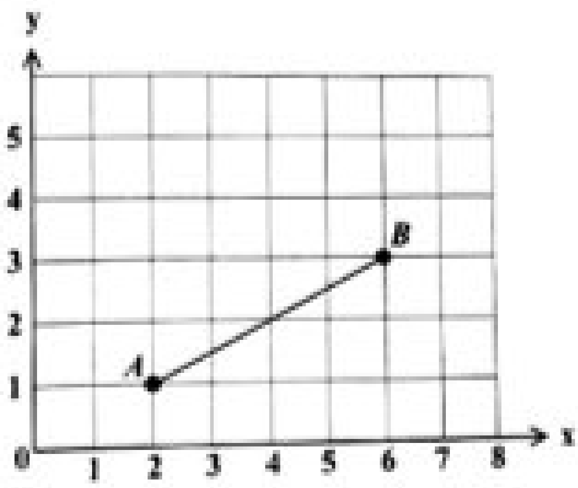
Item 47 refers to the following diagram of a number line.




47. The graph of the inequality in the diagram is defined by

(A) $x < 1$
(B) $x > 1$
(C) $x \geq 1$
(D) $x \leq 1$

Item 48 refers to the following graph of a straight line.




48. The gradient of $AB$ in the graph above is

(A) $-2$

(B) $2$

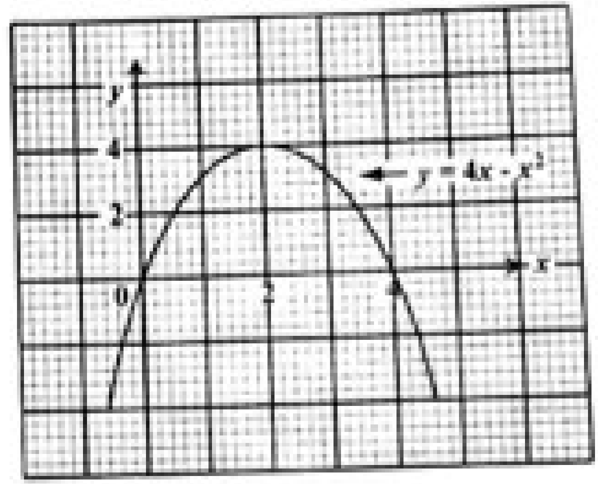
(C) $\frac{-1}{2}$

(D) $\frac{1}{2}$

49. If $f(x) = 2x^2 - 1$, then $f(-3) =$

(A) $-32$
(B) $-19$
(C) $17$
(D) $35$

Item 50 refers to the following graph of a quadratic function.




50. The maximum point of $y = 4x - x^2$ is

(A) $(4, 4)$
(B) $(0, 4)$
(C) $(4, 2)$
(D) $(2, 4)$

51. What is the gradient of the straight line $2y = -3x - 8$?

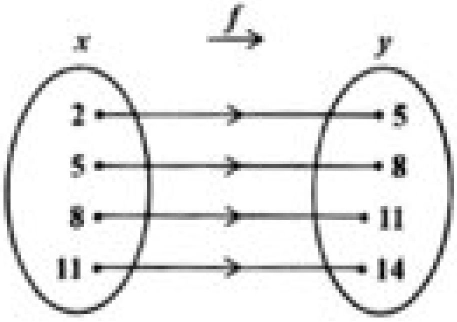
(A) $-3$

(B) $\frac{-3}{2}$

(C) $2$

(D) $3$

Item 52 refers to the following arrow diagram which shows a function, $f$.



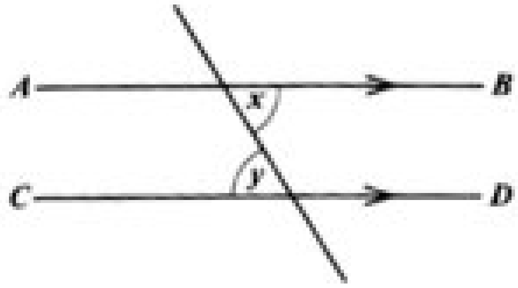

52. Which of the following BEST describes the function?

(A) $y = x + 3$
(B) $x + y = 3$
(C) $x = y + 3$
(D) $y = 2x + 1$

53. If each exterior angle of a regular polygon is $60°$, then the polygon is a

(A) quadrilateral
(B) pentagon
(C) hexagon
(D) triangle

Item 54 refers to the following diagram.




54. In the diagram, $AB$ and $CD$ are parallel. Which of the following BEST describes the relation between $x$ and $y$?

(A) $x = y$
(B) $x > y$
(C) $x + y > 2x$
(D) $x + y < 2x$

Item 55 refers to the following diagram of an isosceles triangle.

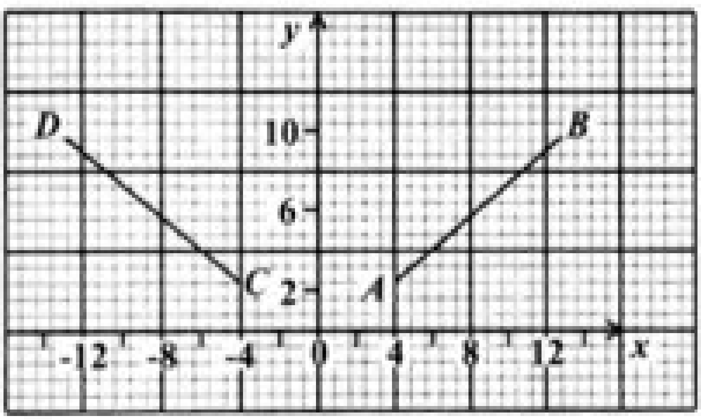

55. In the triangle, the value of $x$ is

(A) $30°$
(B) $60°$
(C) $120°$
(D) $150°$

56. The image of a point $P(-2, 3)$ under a translation $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ is

(A) $(-6, 12)$
(B) $(-5, -1)$
(C) $(5, 1)$
(D) $(1, 7)$

Item 57 refers to the following diagram of two straight lines.




57. In the diagram above, the line $CD$ is the image of $AB$ after

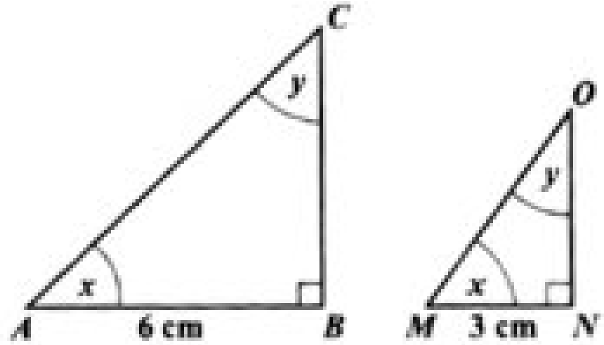
(A) a rotation through 90° centre $O$

(B) a reflection in the $y$-axis

(C) a translation by vector $\begin{bmatrix} -4 \\ -8 \end{bmatrix}$

(D) a reflection in the $x$-axis

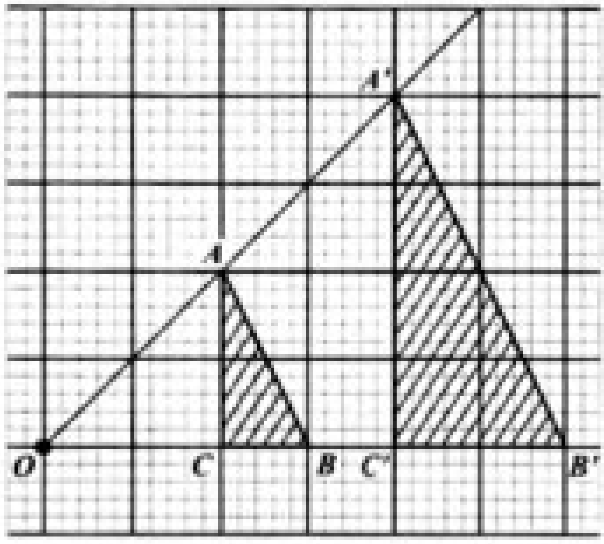
Item 58 refers to the following pair of similar triangles.




58. If the area of $\Delta ABC$ is 20 cm², what is the area of $\Delta MNO$, in cm²?

(A) 2.5
(B) 3.3
(C) 5
(D) 10

Item 59 refers to the following diagram which shows an enlargement.




59. $OAA'$, $OBB'$ and $OCC'$ are straight lines. $\Delta\ ABC$ is mapped onto $\Delta\ A'B'C'$ by an enlargement with centre $O$. What is the scale factor of the enlargement?

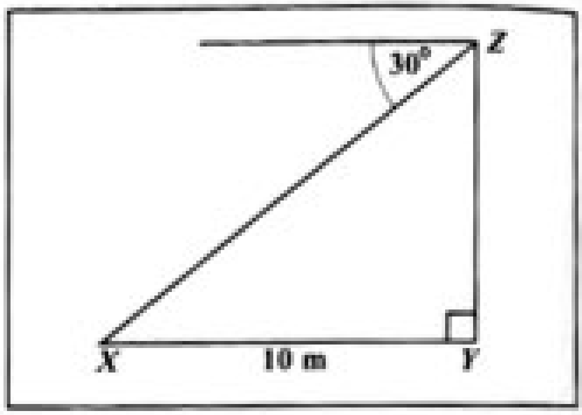
(A) $\frac{1}{2}$

(B) $\frac{-1}{2}$

(C) 2

(D) $-2$

Item 60 refers to the following diagram.




60. The diagram above, **not drawn to scale**, shows that the angle of depression of a point $X$ from $Z$ is 30°. If $X$ is 10 metres from $Y$, the height of $YZ$, in metres, is

(A) 10 tan 30°
(B) 10 sin 30°
(C) 10 cos 30°
(D) 10 cos 60°

**END OF TEST**

**IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.**